

LIBBY CASE REPORT

osteosarcoma-mets-dll3-h7r2

Multi-agent decision support — clinician track

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Libby is an experimental decision-support tool. The recommendations in this report have not been reviewed by a clinician treating this patient. Do not act on this report without consulting a qualified oncologist.

Executive summary

Case: osteosarcoma-mets-dll3-h7r2 **Question:** Metastatic osteosarcoma with user-reported DLL3 RNA expression. Are DLL3-directed therapies a reasonable option, and what should the patient pursue if they aren't? **Evidence base:** 5 trials, 5 clinical-evidence rows, 4 preclinical rows. Board agreement ranged from full consensus on regorafenib and the DLL3 IHC workup down to one persistent dissent on the lead trial option.

What this report covers

The PI's synthesis across two parallel ranked tracks: one assumes DLL3 protein confirmation by IHC, the other assumes the IHC comes back negative. The shared first action, IHC SP347 on tumor, is rank 1 under both branches.

Top-line findings

- DLL3 IHC is the test the rest of the case turns on. RNA expression alone does not establish membrane DLL3, and every DLL3-directed therapy in current clinical use requires IHC confirmation. All five personas converged on this without dissent.
- In the IHC-positive branch, the lead option is tarlatamab via NCT06788938, a basket trial that includes osteosarcoma. The conservative's earlier toxicity veto lifts on biomarker confirmation.
- The critic's dissent on the trial persists in BOTH scenarios: there are no published osteosarcoma data with tarlatamab, and cross-tumor translation from SCLC is unproven. That's the scientific question this enrollment would actually answer.
- Regorafenib (two RCTs) has unanimous board endorsement in both scenarios. In the IHC-negative branch it ranks first because the trial pathway closes.

Recommendation summary

Shared first step: DLL3 IHC (SP347) on tumor, gates branch selection. All 5 personas endorse.

Path A, if DLL3 IHC $\geq 1\%$ (preferably $\geq 25\%$):

1. **Tarlatamab via NCT06788938**— strong preference fit (high-risk-high-reward + prefers trials); critic dissent on cross-tumor translatability persists.
2. **Regorafenib**— off-trial backbone; SARC024 + REGOBONE evidence base; universal endorsement.
3. **Cabozantinib**— NCCN alternative; CABONE phase-2 ORR 11.9%.

Path B, if DLL3 IHC negative or below threshold:

1. **Regorafenib**— universal endorsement; two RCTs; strongest evidence among the remaining options.
2. **Cabozantinib**— sequencing alternative.
3. **Non-DLL3 trial search (mechanism-agnostic)**— considered with caveats; honors the prefers_trials preference even with the DLL3 pathway closed.

What this report does **not** cover

Dose adjustments, monitoring schedules, sequencing across lines, payer/access logistics, or institution-specific trial slot availability. Those belong to the treating team.

How to use this report

Each ranked option carries the board's agreement state (endorsed, dissent, or veto by persona) and 1–3 anchor citations. The full per-trial extraction lives on the live case page; this PDF is the synthesis. Both scenario tracks run in parallel, so the patient and clinician can plan for either IHC result without waiting for the test to come back.

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<!-- libby:downloads:begin -->

Downloads

- [Clinician PDF report](#)— ranked recommendations + evidence + sources
- [Patient/caregiver PDF](#)— plain-language summary
- [Self-contained HTML](#)— recommendations table that opens offline

<!-- libby:downloads:end -->

Profile snapshot (scrubbed)

- **Primary site / histology:**bone — osteosarcoma
- **Stage:**IV (metastatic)
- **Performance status (assumed):**ECOG 1
- **Age band (assumed):**18-29 (typical osteosarcoma demographics; not user-supplied)
- **Sex:**unknown
- **Biomarkers:****DLL3 — RNA only (confirmation_status: rna_only); IHC SP347 status unknown.**
Decision-relevant resolution: ≥1% (preferably ≥25%) by IHC for DLL3-directed clinical trials.
- **Prior therapy (assumed):**MAP frontline; response not provided

Preferences

- **Efficacy/toxicity weight:**0.85 (strong efficacy lean)
- **Toxicity vetoes:**none stated
- **Modality constraints:**none stated
- **Free text:**"accepts high-risk high-reward options"
- **Trials preferred:**yes

The scenario fork

Because DLL3 is *rna_only* and DLL3-directed therapies require IHC protein confirmation, the PI emits **two parallel ranked tracks** below — the recommendations the user can plan against under each branch of the IHC result. The single shared first action is the IHC test itself.

Shared first step (applies to every scenario)

Rank	Status	Action	Endorsed
1	recommended	DLL3 IHC (SP347) on tumor — gates branch selection	all 5 personas

The IHC is non-toxic, takes 1–3 weeks, and is required for trial NCT06788938 regardless of which branch the result places the patient in.

Path A — if DLL3 IHC ≥1% (preferably ≥25%)

Rank	Status	Intervention	Endorsed	Dissent	Veto
1	recommended	tarlatamab via NCT06788938	risktaker, advocate, conservative, consensusite	critic	—
2	recommended	regorafenib (off-trial backbone)	all 5	—	—
3	recommended	cabozantinib (alternative)	all 5	—	—

In the positive scenario, the **conservative's earlier toxicity veto on tarlatamab lifts**— its own comment specified the veto was conditional on IHC confirmation. Consensusite's qualified-on-guideline-fit critique upgrades to endorsement once the trial-enrollment principle and biomarker-fit principle align. Risktaker and advocate continue to endorse from round 1. The critic's **evidence-quality dissent persists**— there are no published osteosarcoma data with tarlatamab; cross-tumor translation from SCLC remains unproven and is the load-bearing scientific question this trial enrollment would answer.

The user's stated preferences (high-risk-high-reward + prefers_trials) point exactly here.

Path B — if DLL3 IHC negative or below threshold

Rank	Status	Intervention	Endorsed	Dissent	Veto
1	recommended	regorafenib	all 5	—	—
2	recommended	cabozantinib	all 5	—	—
3	considered with caveats	non-DLL3 trial search (mechanism-agnostic)	risktaker, advocate	—	—

In the negative scenario, the tarlatamab pathway is foreclosed. The advocate's prior rank-disagreement on regorafenib (which was a rank-issue, not a drug-issue) goes away because the alternative they were championing is no longer available. Regorafenib's two-RCT evidence base ([Davis 2019 SARC024](#), [Duffaud 2019 REGOBONE](#)) is the highest-utility option remaining. The third row honors the user's prefers_trials preference even with the DLL3 pathway closed — a referral to an academic sarcoma center for a fresh trial search is the residual high-EV move.

What was NOT confirmed

The user-supplied "DLL3 RNA expression" was not converted into actionable biomarker status. The board's strongest single signal — heard from all five personas — is that this conversion (RNA → IHC protein) is the next step in care planning. The two ranked tracks above let the user plan for both results in parallel rather than serially.

Transparency artifacts

- [Trial table](#)— 5 trials in the dossier

- [Evidence list](#)— 5 clinical-evidence rows + 4 preclinical rows
- [Tumor-board transcript](#)— 5 positions, 20 cross-critiques
- [Recommendations table](#)— full ranked detail with scenario branches
- [Plain-language summary](#)— patient/caregiver track

Run log

Authored May 2026. Inputs supplied: targetable feature ("DLL3 RNA expression"), clinical descriptor ("metastatic osteosarcoma"), preference summary ("accepts high-risk high-reward options"). Re-run after Libby gained scenario-branching support for non-confirmed biomarkers.

Sources

PubMed (PMID):

- [35983951](#)
- [37861218](#)
- [40454646](#)
- [31013172](#)
- [30477937](#)
- [32078813](#)

ClinicalTrials.gov (NCT):

- [NCT06788938](#)